MIL-S-19500/203C(NAVY)
9 March 1970
SUPERSEDING
MIL-S-19500/203B(NAVY)
7 June 1968

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS) SILICON TYPES 2N2024, 2N2025, 2N2027, 2N2029, 2N2030

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a reverse blocking triode thyristor, silicon types 2N2024, 2N2025, 2N2027, 2N2029, and 2N2030, and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Physical dimensions. See figure 2.

1.3 Maximum ratings.

	I_O 1/	$i_{FM(surge)}$ 2/ 1 cycle	V_{KG}	v_{GKM}	i_{GKM}	P_{GM}	$P_{G(av)}$	Stud torque	Barometric pressure
	A	a	V	v	a	w	w	in.lbs	mm Hg
Minimum	---	----	--	--	--	--	--	---	15
Maximum	70	1000	5	10	2	5	0.5	150	---

1/ This average forward current is for a maximum allowable case temperature of 87° C and 180 electrical degrees of half sine wave conduction. For other maximum operating conditions see figure 1.
2/ Surge rating is nonrecurrent and implies only with device in conducting state.

OPERATING TEMPERATURE: -40° C to +150° C
STORAGE TEMPERATURE: -40° C to +150° C

1.3.1 Individual ratings.

	V_{RM} 1/	V_{FBOM}	$V_{RM(Non-rep)}$ 2/
	V	V	V
2N 2024	50	50	75
2N 2025	100	100	150
2N 2027	200	200	300
2N 2029	300	300	400
2N 2030	400	400	500

1/ Values apply for zero or negative gate voltage (V_{KG}).
2/ Transient inverse voltage, nonrecurrent, t=5.0 msec max.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

1.4 Characteristics.

	V_{FM}	I_{HOO}	V_{GT} 1/	I_{GT} 1/	Critical dv/dt 1/
	v	mAdc	V_{dc}	mAdc	v/usec
Min	---	---	0.25	---	20
Max	1.6	40	3.00	130	---

1/ T_A = -40° to +150° C

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component parts
MIL-STD-750 - Test Methods for Semiconductor Devices

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements for the thyristors shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500 and as follows:

V_{GG} - Gate power supply voltage (dc)

V_{AA} - Anode power supply voltage (dc)

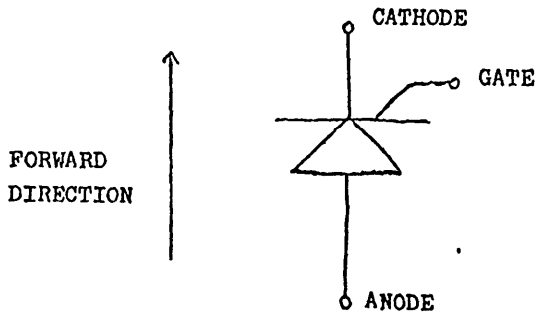
V_{FBX} - Forward blocking voltage, gate connected to the specified reference terminal

3.3 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Country of origin

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

3.3.1 Polarity marking on controlled power rectifiers. The polarity shall be indicated by marking the body with the graphic symbol for thyristors so oriented that the arrow points in the direction of easy flow of the conventional current (forward direction).



3.4 Design, construction, and physical dimensions. Thyristors shall be of the design, construction, and physical dimensions shown on figure 2.

3.5 Performance characteristics. The performance characteristics of thyristors shall be as specified in tables I, II, and III.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein. The devices covered by this detail specification shall be considered to be structurally similar for the purpose of inspection.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.2.1 Subgroups 1 and 4 of group B inspection and subgroup 5 of group C inspection shall be performed on any structurally similar type on a sublot basis to qualify that type and all structurally similar lower-voltage types.

4.2.2 Subgroup 2 of group B inspection shall be performed on a full sample for each type being qualified.

4.2.3 Subgroup 3 of group B inspection and subgroups 1, 2, and 4 of group C inspection may be performed in accordance with MIL-S-19500 or may be performed on a full sample of any type to qualify all types.

4.2.4 Subgroup 3 of group C shall be performed on a full sample for the highest voltage type to qualify it and all lower-voltage types tested at the same barometric pressure condition.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections. When specified in the contract or order, one copy of the quality conformance inspection data, pertinent to the device inspection lot, shall be supplied with each shipment by the device manufacturer. (See 6.2).

4.3.1 Group A inspection. Group A inspection shall consist of examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. Group C inspection shall be conducted on the first lot every 6 months. Successful completion of the group C inspection shall permit acceptance of lots for the period specified on the basis of completion of group A and group B inspection. If a group C failure occurs, inspection of the failed subgroup shall revert to a lot basis until three consecutive lots have passed.

4.3.4 Inspection procedure.

4.3.4.1 Subgroups 1 and 2 of group B inspection shall be performed on a full sample size of each type in the lot.

4.3.4.2 Subgroup 3 of group B inspection and subgroups 1, 2, and 4 of group C inspection may be performed in accordance with MIL-S-19500 or may be performed on any type within the lot to accept all types in the lot.

4.3.4.3 Subgroup 4 of group B inspection shall be performed on a full sample size of the highest voltage type represented in the lot to accept all types in the lot.

4.3.4.4 Subgroups 3 and 5 of group C inspection shall be performed on a full sample size of the highest voltage type represented in the lot to accept that voltage type and all lower-voltage types for the specified period. Subsequent acceptance of voltage types which are higher than previously subjected to subgroups 3 and 5 of group C, within the current specified period, requires retesting of subgroups 3 and 5 of group C for a lot including the higher-voltage type. Provisions of MIL-S-19500 early acceptance procedures do not apply to this specification.

4.3.4.5 Inspection lot. Inspection lot shall be as defined in MIL-S-19500, except that lot accumulation period requirements shall be 6 months in lieu of 6 weeks.

4.4 Disposition of sample units. Sample units which have been subjected to and have passed subgroups 1, 2, and 4 of group B and subgroups 1, 2, and 5 of group C inspection may be delivered on the contract or order provided that, after the inspection is completed, these sample units are subjected to and pass group A inspection. The tests specified in subgroup 3 of group B are considered destructive.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

4.5.1 Inspection conditions. All measurements shall be made at T_C = 25 ±3° C unless otherwise specified.

4.5.2 End-point mounting torque. The device shall be mounted on a 0.062-inch thick flat copper plate having a mounting-hole diameter of 0.475 inch minimum, 0.500 inch maximum, by means of a 1/2-20 UNF 2A brass nut. The device shall be held by its hex and the specified torque of 15 pound-inches gradually applied to the nut without shock, using a P.A. Sturtevant model F-100-1 torque wrench, or equivalent.

4.5.3 Time limit for end points. End point tests for qualification and quality conformance inspection shall be completed within 96 hours after completion of the last test in the subgroup.

4.5.4 Blocking life test. The specified peak voltage shall be applied from anode to cathode with the gate terminal open circuited. The voltage shall be a 60 Hz sine wave. The cell stud temperature shall be maintained at the specified value throughtout the specified test time. A resistor of 50 ohms shall be connected in series with the voltage power supply to limit current in the event of cell voltage breakdown. The basic circuit shown on figure 4 may be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

4.5.5 Intermittent life test. The controlled power rectifiers shall be operated in a single phase circuit as shown on figure 3 with 175 ±5 degree conduction angle. Cell stud temperature shall be maintained at 100° ±5° C as measured with a thermocouple rigidly attached to the cell hex. Peak input voltage from a 60 Hz sinusoidal blocking voltage power supply shall be equal to the rated blocking voltage. Average forward current shall be 50 amperes per cell from the 60 Hz sinusoidal forward current supply. The blocking voltage supply shall be phase delayed 90 electrical degrees with respect to the forward current supply in order to impose both forward and reverse blocking duty on the cells under test. The forward current supply consists of a low voltage transformer with a means of adjusting the secondary voltage. The blocking voltage supply consists of a high voltage transformer with a synchronous switch connected in series with one of the secondary transformer leads. The synchronous switch is set to conduct for 170 ±5 electrical degrees conduction during the interval when terminal A is negative with respect to terminal C. A minimum of 2 and a maximum of 5 electrical degrees off time shall be allowed after forward current conduction before blocking voltage is applied by the synchronous switch. The gate firing pulses shall be delivered from a firing circuit with an open circuit voltage of 7.0 ±3.0 volts and a short circuit of 1.2 ±0.8 amperes. The gate pulse average power input shall not exceed 0.5 watt. The gate firing supply shall be in phase with the cell forward current supply.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical inspection	2071			---	---	---	---
Subgroup 2			5				
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz		i_{RBOM}			
2N2024		v_{RM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{RM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{RM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{RM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{RM} = 400 v (pk)			---	15	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz		i_{FBOM}			
2N2024		v_{FBOM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{FBOM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{FBOM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{FBOM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{FBOM} = 400 v (pk)			---	15	ma(pk)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection (cont'd)

Examination or test	MIL-STD-750 Method	MIL-STD-750 Details	LTPD	Symbol	Limits Min	Limits Max	Unit
Subgroup 3			5				
High temperature reverse blocking current	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz		i_{RBOM}			
2N2024		v_{RM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{RM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{RM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{RM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{RM} = 400 v (pk)			---	15	ma(pk)
High temperature forward blocking current	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz		i_{FBOM}			
2N2024		v_{FBOM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{FBOM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{FBOM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{FBOM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{FBOM} = 400 v (pk)			---	15	ma(pk)
Exponential rate of voltage rise	4231	Bias cond. D; T_C = 150° C; $\frac{dv}{dt}$ = 20 v/usec; repetition rate = 60 pps; test duration = 15 sec; C = 1.0 uf; 10Ω ≤ R_L ≤50Ω					
2N2024		V_{AA} = 50 Vdc			47	---	Vdc
2N2025		V_{AA} = 100 Vdc			95	---	Vdc
2N2027		V_{AA} = 200 Vdc			190	---	Vdc
2N2029		V_{AA} = 300 Vdc			285	---	Vdc
2N2030		V_{AA} = 400 Vdc			380	---	Vdc
Critical gate trigger voltage	4221	T_C = 150° C; R_L = 10Ω; R_e = 25Ω max		V_{GT}			
2N2024		$V_2 = V_{FBX}$ = 50 Vdc			0.25	---	Vdc
2N2025		$V_2 = V_{FBX}$ = 100 Vdc			0.25	---	Vdc
2N2027		$V_2 = V_{FBX}$ = 200 Vdc			0.25	---	Vdc
2N2029		$V_2 = V_{FBX}$ = 300 Vdc			0.25	---	Vdc
2N2030		$V_2 = V_{FBX}$ = 400 Vdc			0.25	---	Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 4			5				
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6.0 Vdc; R_L = 10Ω max; R_e = 25Ω max		V_{GT}	---	3.0	Vdc
				I_{GT}	---	70.	mAdc
Forward "on" voltage	4226	i_{FM} = 220a(pk) (pulse); pulse width = 8.5 msec. max; duty cycle = 2% max		v_{FM}	---	1.75	v(pk)
Holding current	4201	Bias cond. D; V_{AA} = 22.5 Vdc; I_{F1} = 5 Adc; I_{F2} = 100 mAdc; trigger voltage source = 6 Vdc; trigger PW = 25 usec (min); R_2 = 50Ω		I_{HOO}	---	40	mAdc
Turn-off time	---	(See figures 5a and 5b) i_{FM} = 70 a(pk) t_p = 100 ±50 usec; di/dt = 5 a/usec, min; reverse voltage at t_1 = 2 volts, min; T_C = 145° C; repetition rate = 60 pps, max; dv/dt = 20 v/usec; gate bias conditions: gate source voltage = 0.0 volts; gate source resistance = 100 ohms		t_{off}			
2N2024		$v_{FBXM} = v_{FBOM}$ = 50 v(pk); v_{RM} = 50 v, max			---	150	usec

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 4 (cont'd)							
2N2025		$V_{FBXM} = V_{FBOM} =$ 100 v(pk); V_{RM} = 100 v, max			---	150	usec
2N2027		$V_{FBXM} = V_{FBOM} =$ 200 v(pk); V_{RM} = 200 v, max			---	150	usec
2N2029		$V_{FBXM} = V_{FBOM} =$ 300 v(pk); V_{RM} = 300 v, max			---	150	usec
2N2030		$V_{FBXM} = V_{FBOM} =$ 400 v(pk); V_{RM} = 400 v, max			---	150	usec
Gate-controlled turn-on time	4223	i_{FM} = 70 Adc; V_{GG} = 10 Vdc; R_e = 25Ω; t_{P1} = 20 ±5 usec, 2 a/usec $\leq \frac{di}{dt} \leq$ 100 a/usec		t_{on}	---	5	usec
2N2024		V_{AA} = 50 Vdc					
2N2025		V_{AA} = 100 Vdc					
2N2027		V_{AA} = 100 Vdc					
2N2029		V_{AA} = 100 Vdc					
2N2030		V_{AA} = 100 Vdc					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Reverse gate current	4219	V_{KG} = 5 Vdc		I_{KG}	---	300	mAdc
Surge current	4066	i_{FM} (surge) = 1000a; 10 surges at 1 per minute; T_C = 150° C; f = 60 Hz; surge duration = 7 msec, min, rated V_{RM}			---	---	---
Nonrepetitive peak reverse voltage	---	(See figure 6); 10 pulses		---	---	---	---
2N2024		v_{RM} = 75 v (pk)					
2N2025		v_{RM} = 150 v (pk)					
2N2027		v_{RM} = 300 v (pk)					
2N2029		v_{RM} = 400 v (pk)					
2N2030		v_{RM} = 500 v (pk)					
End points: (see 4.5.3)							
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz		i_{RBOM}			
2N2024		v_{RM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{RM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{RM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{RM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{RM} = 400 v (pk)			---	15	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz		i_{FBOM}			
2N2024		v_{FBOM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{FBOM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{FBOM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{FBOM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{FBOM} = 400 v (pk)			---	15	ma(pk)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE II. Group B inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1 (cont'd.)							
Reverse blocking current at high temperature	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz		i_{RBOM}			
2N2024		v_{RM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{RM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{RM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{RM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{RM} = 400 v (pk)			---	15	ma(pk)
Forward blocking current at high temperature	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz		i_{FBOM}			
2N2024		v_{FBOM} = 50 v (pk)			---	15	ma(pk)
2N2025		v_{FBOM} = 100 v (pk)			---	15	ma(pk)
2N2027		v_{FBOM} = 200 v (pk)			---	15	ma(pk)
2N2029		v_{FBOM} = 300 v (pk)			---	15	ma(pk)
2N2030		v_{FBOM} = 400 v (pk)			---	15	ma(pk)
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6.0 Vdc; R_L = 10Ω, R_e = 25Ω, max		V_{GT}	---	3.0	Vdc
				I_{GT}	---	70	mAdc
Forward "on" voltage	4226	i_{FM} = 200a (pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2%, max		v_{FM}	---	1.60	v(pk)
Subgroup 2			10				
Low temperature reverse blocking current	4211	AC method; bias cond. D; T_C = -40° C; f = 60 Hz		i_{RBOM}			

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE II. Group B inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 (cont'd)							
2N2024		V_{RM} = 50 v (pk)			---	15	ma(pk)
2N2025		V_{RM} = 100 v (pk)			---	15	ma(pk)
2N2027		V_{RM} = 200 v (pk)			---	15	ma(pk)
2N2029		V_{RM} = 300 v (pk)			---	15	ma(pk)
2N2030		V_{RM} = 400 v (pk)			---	15	ma(pk)
Low temperature forward blocking current	4206	AC method; bias cond. D; T_C = -40° C; f = 60 Hz		i_{FBOM}			
2N2024		V_{FBOM} = 50 v (pk)			---	15	ma(pk)
2N2025		V_{FBOM} = 100 v (pk)			---	15	ma(pk)
2N2027		V_{FBOM} = 200 v (pk)			---	15	ma(pk)
2N2029		V_{FBOM} = 300 v (pk)			---	15	ma(pk)
2N2030		V_{FBOM} = 400 v (pk)			---	15	ma(pk)
Low temperature gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6.0 Vdc; T_C = -40° C; R_L = 10Ω, R_e = 25Ω, max		V_{GT} I_{GT}	--- ---	3.0 130	Vdc mAdc
Subgroup 3			15				
Thermal shock (temperature cycling)	1051	Test cond F, except T_{min} = -40° C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. B		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points: (Same as subgroup 1)							
Subgroup 4			10				
Blocking life test (See 4.5.4)	---	(See figure 4) T_C = 147° C; t = 500 hrs.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE II. Group B inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 4 (cont'd)							
2N2024		$V_{FBOM} = V_{RM}$ = 50 v (pk)		---	---	---	---
2N2025		$V_{FBOM} = V_{RM}$ = 100 v (pk)		---	---	---	---
2N2027		$V_{FBOM} = V_{RM}$ = 200 v (pk)		---	---	---	---
2N2029		$V_{FBOM} = V_{RM}$ = 300 v (pk)		---	---	---	---
2N2030		$V_{FBOM} = V_{RM}$ = 400 v (pk)		---	---	---	---

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 2)		---	---	---	---
Subgroup 2			10				
Shock	2016	Nonoperating; 500 G, 1.0 msec; 5 blows each in orientation; X_1, Y_1, and Y_2		---	---	---	---
Vibration, variable frequency	2056	Nonoperating 5 G (leads removed or secured)		---	---	---	---
Constant acceleration	2006	2,500 G; in orientations X_1, Y_1, and Y_2 (leads secured or removed)		---	---	---	---
Terminal strength (stud torque) (See 4.5.2)	2036	Test cond. D2 150 in. lbs.		---	---	---	---
End points: (see 4.5.3)							
Reverse blocking current	4211	AC method; test cond. D; f = 60 Hz		i_{RBOM}			

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE III. Group C inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 (cont'd)							
2N2024		V_{RM} = 50 v (pk)			---	15	ma(pk)
2N2025		V_{RM} = 100 v (pk)			---	15	ma(pk)
2N2027		V_{RM} = 200 v (pk)			---	15	ma(pk)
2N2029		V_{RM} = 300 v (pk)			---	15	ma(pk)
2N2030		V_{RM} = 400 v (pk)			---	15	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz		i_{FBOM}			
2N2024		V_{FBOM} = 50 v (pk)			---	15	ma(pk)
2N2025		V_{FBOM} = 100 v (pk)			---	15	ma(pk)
2N2027		V_{FBOM} = 200 v (pk)			---	15	ma(pk)
2N2029		V_{FBOM} = 300 v (pk)			---	15	ma(pk)
2N2030		V_{FBOM} = 400 v (pk)			---	15	ma(pk)
Reverse blocking current at high temperature	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz		i_{RBOM}			
2N2024		V_{RM} = 50 v (pk)			---	15	ma(pk)
2N2025		V_{RM} = 100 v (pk)			---	15	ma(pk)
2N2027		V_{RM} = 200 v (pk)			---	15	ma(pk)
2N2029		V_{RM} = 300 v (pk)			---	15	ma(pk)
2N2030		V_{RM} = 400 v (pk)			---	15	ma(pk)
Forward blocking current at high temperature	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz		i_{FBOM}			
2N2024		V_{FBOM} = 50 v (pk)			---	15	ma(pk)
2N2025		V_{FBOM} = 100 v (pk)			---	15	ma(pk)
2N2027		V_{FBOM} = 200 v (pk)			---	15	ma(pk)
2N2029		V_{FBOM} = 300 v (pk)			---	15	ma(pk)
2N2030		V_{FBOM} = 400 v (pk)			---	15	ma(pk)
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6.0 Vdc; R_L = 10Ω max, R_e = 25Ω, max		V_{GT} I_{GT}	--- ---	3.0 70	Vdc mAdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE III. Group C inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 (cont'd)							
Forward "on" voltage	4226	i_{FM} = 200a (pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2% max		V_{FM}	---	1.6	v(pk)
Subgroup 3			15				
Barometric pressure, reduced	1001	t = 60 seconds forward blocking and 60 seconds reverse blocking; pressure; 15 mm Hg					
		Voltage applied during test:					
2N2024		$V_{FBO} = V_R$ = 50 Vdc		---	---	---	---
2N2025		$V_{FBO} = V_R$ = 100 Vdc		---	---	---	---
2N2027		$V_{FBO} = V_R$ = 200 Vdc		---	---	---	---
2N2029		$V_{FBO} = V_R$ = 300 Vdc		---	---	---	---
2N2030		$V_{FBO} = V_R$ = 400 Vdc		---	---	---	---
Subgroup 4			15				
Salt atmosphere (corrosion)	1041			---	---	---	---
Subgroup 5			λ = 20				
Intermittent life (See 4.5.5)		50 min. on 10 min. off		--- ---	--- ---	--- ---	--- ---

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

TABLE III. Group C inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 5 (cont'd) End points: (Same as sub-group 2 except when the 25° C forward and reverse blocking current tests are performed, the device shall be mounted in accordance with 4.5.2)							

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data:

(a) Inspection date (see 4.3).

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

(Project 5961-N299)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

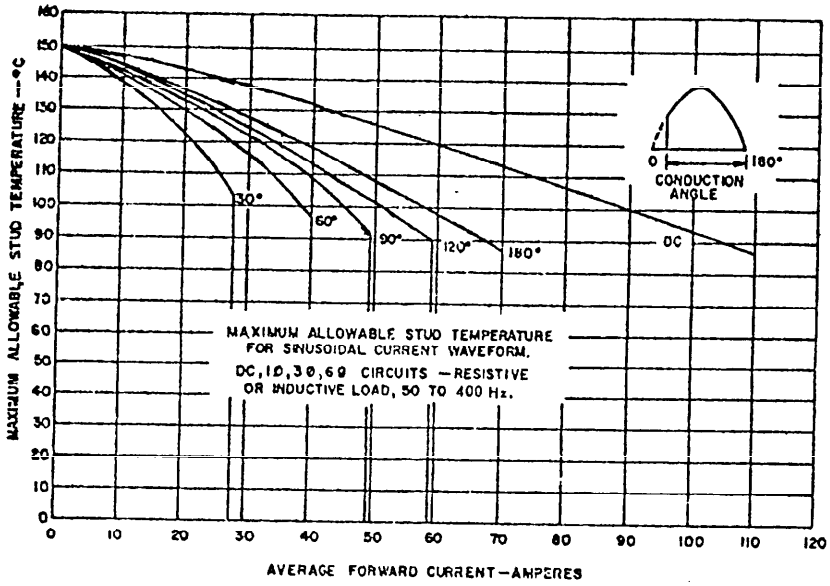
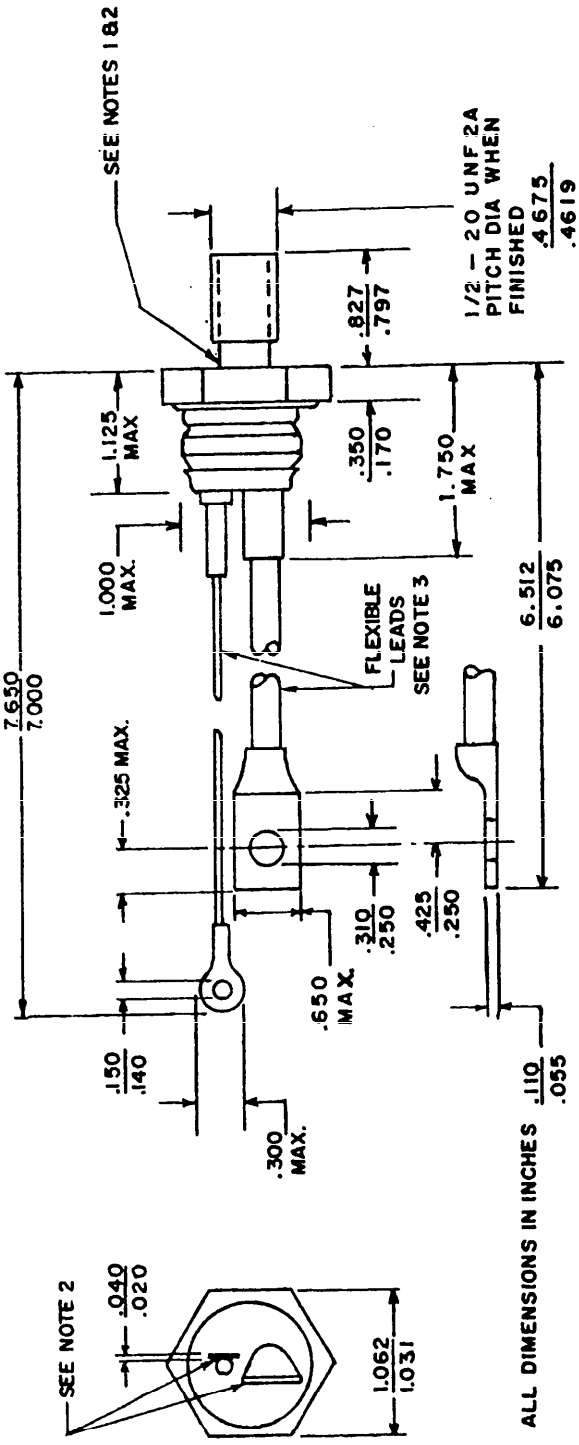


FIGURE 1. Maximum allowable stud temperature.

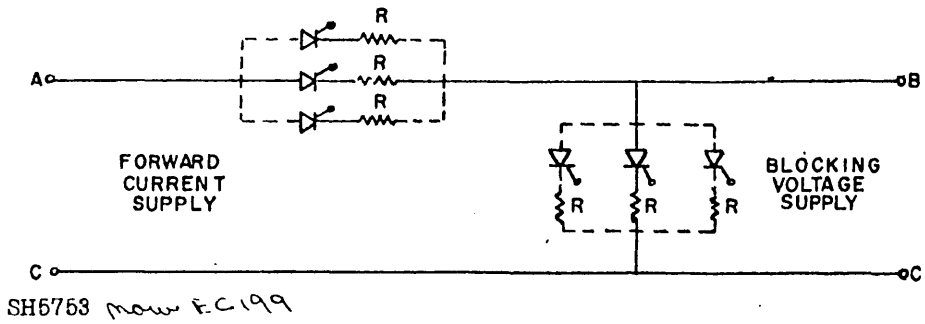
Source: https://assist.dla.mil -- Downloaded: 202610-18T13:40Z
Check the source to verify that this is the current version before use.



NOTES:
1. COMPLETE THREADS TO EXTEND TO WITHIN 2-1/2 THREADS OF HEAD.
2. ANGULAR ORIENTATION OF THESE TERMINALS IS UNDEFINED.
3. SILICON RUBBER INSULATION (CLASS H), 300 VOLT ASTM STANDARD WALL.

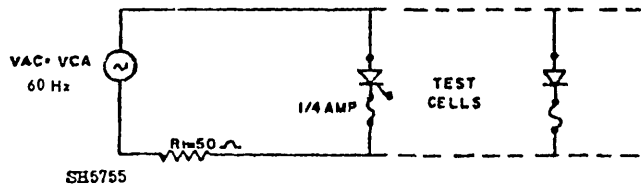
FIGURE 2. Dimensions of thyristor types 2N2024, 2N2025, 2N2027, 2N2029, and 2N2030.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.



RESISTOR R IS FOR EQUAL CURRENT DISTRIBUTION BETWEEN PARALLELED CELLS, APPROXIMATELY 0.005 OHM.

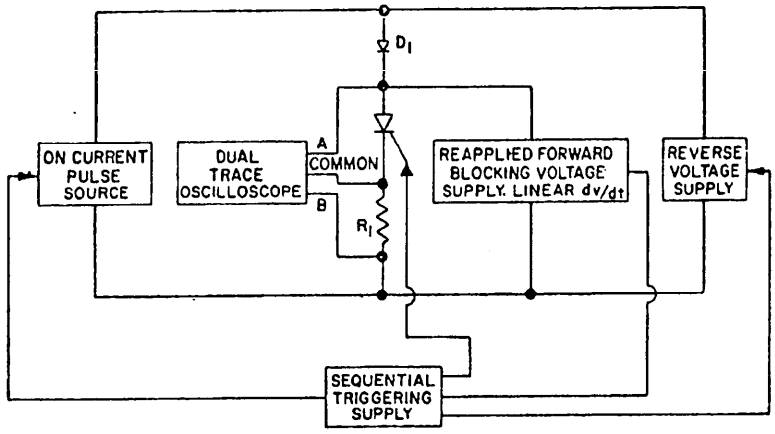
FIGURE 3. Basic test circuit for intermittent life test.



Note 1: A 1/4 amp fuse shall also be connected in series with each cell to serve as an indicator of cell voltage breakdown. Failure shall be indicated by either a blown fuse or failure to meet the specified end point test limits. The equipment shall be designed to limit peak transient voltage values to the cell voltage rating ($V_{AC} = V_{CA}$).

FIGURE 4. Basic circuit for blocking life test.

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:40Z
Check the source to verify that this is the current version before use.



NOTES:
1. R_1 is a non-inductive shunt.
2. D_1 must be selected for recovery time longer than the recovery time of the device under test, but shorter than the turn-off time of the device under test.

FIGURE 5a. Test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

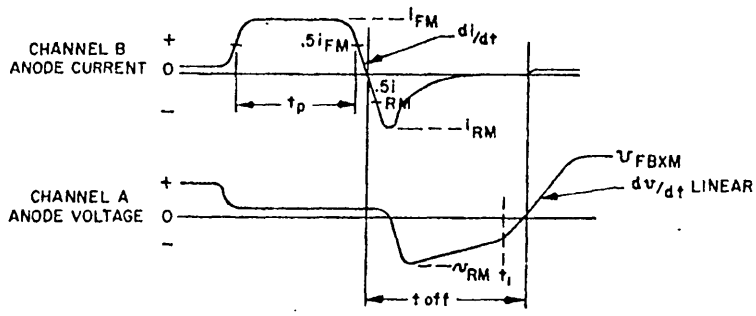


FIGURE 5b. Waveforms.

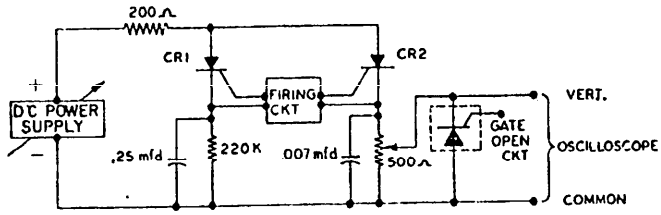
Procedure: The on-current pulse source is adjusted to apply current through the device under test of the specified magnitude, pulse width and duty cycle. The reverse voltage supply is adjusted to apply reverse current to the device under test with the specified di/dt and specified peak reverse voltage and reverse voltage at time t_1. The reapplied forward blocking supply is adjusted to apply forward voltage to the device under test of the specified value with the specified linear rate of rise. The time between when the current passed through zero until the reapplied voltage passes through zero is decreased until either the device under test fails to block the reapplied voltage, or the specified t_{off} is reached. The turn-off time is read from the oscilloscope as shown on figure 5b.

*Operation conditions: The devices shall be subjected to the following operational conditions:

a. Forward current amplitude, i_{FM}
b. Forward current duration, t_p
c. Commutation rate, di/dt (the slope of the line from 50% of + peak to 50% of – peak)
d. Peak reverse voltage (max), v_{RM}
e. Reverse voltage at t_1 (min), v_R at t_1
f. Test temperature
g. Rate of rise of reapplied forward voltage, dv/dt
h. Forward blocking voltage, v_{FBXM}
i. Gate bias condition (between gate trigger pulses):
   (1) Gate source voltage
   (2) Gate source resistance
j. Repetition rate

FIGURE 5. Test circuit and waveforms for turn-off time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.



Procedure: Adjust potentiometer to achieve the specified voltage across the device under test. Timing sequence of firing circuit is: apply trigger pulse to CR2 once every 15 seconds, apply trigger pulse to CR1, 5.0±1.0 milliseconds after firing CR2. The specified number of pulses shall be applied. Alternate circuits shall apply rectangular pulses of 5.0±1.0 milliseconds with a maximum rise and fall time of 50 microseconds.

FIGURE 6. Non-repetitive peak reverse voltage test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1800

☆ U.S. GOVERNMENT PRINTING OFFICE 1973—714-666/3983 2-1

Source: https://assist.dla.mil -- Downloaded: 20260-18T13:40Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Naval Electronic Systems Command
Attn: ELEX 5043
Department of the Navy
Washington, DC 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:40Z
Check the source to verify that this is the current version before use.